

*Effectiveness of dapagliflozin and sitagliptin fixed dose combination in Indian patients with type 2 diabetes mellitus and Cardiovascular Comorbidities – Subset analysis of a Real-World Retrospective Study from Electronic Medical Records (EMR)*



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Type 2 diabetes mellitus (T2DM) is a chronic metabolic disorder that is prevalent in India. The estimated prevalence of T2DM in **Indian adults is 11.4%**. In 2021, it was **estimated that 101 million individuals in India** had diabetes<sup>1</sup>

**Cardiovascular comorbidities** like ischemic heart disease, ischemic cerebrovascular disease, or peripheral artery disease are **frequently observed in diabetic patients and leading to morbidity and mortality**<sup>2</sup>

**Managing both diabetes and cardiovascular conditions is crucial to reduce the risk of these life-threatening complications** and improve patient outcomes<sup>2</sup>

**Diabetes is a progressive disease**, treatment intensification with **multiple oral anti-diabetic agents** is often necessary. However, **complex treatment regimens** can lead to **poor adherence and inadequate glycemic control**.<sup>3,4</sup>

**The use of a fixed-dose combination (FDC) of agents with complementary mechanisms of action, like FDC of dapagliflozin and sitagliptin, has shown to improve patient compliance, treatment adherence, and glycemic control**<sup>3</sup>

- The combination of dapagliflozin with sitagliptin has a synergistic effect
- SGLT2 inhibitors increase glucagon concentration and hepatic glucose production while DPP-4 inhibitors stimulate postprandial insulin release and inhibit glucagon release<sup>3,4</sup>

**Dapagliflozin shown to be safe and beneficial in cardiovascular disease (CVD) including heart failure**<sup>5</sup>

Real-world data are important to understand the generalizability and **applicability of FDC of dapagliflozin and sitagliptin, in clinical practice** and to inform clinicians about everyday patient treatment decisions



To date, there is limited real-world data on effectiveness of FDC of dapagliflozin and sitagliptin in India



This is a real-world, **retrospective, observational, multicentre and EMR based study**



**A subset analysis of effectiveness of dapagliflozin and sitagliptin FDC in T2DM patients with Cardiovascular (CV) comorbidities**

**FDCs of drugs with complementary MOAs and  $\beta$ -cell preserving action – can improve compliance, provide effective glycaemic control, lead to weight loss without increased risk of hypoglycaemia <sup>6,7</sup>**

**To evaluate the effectiveness of dapagliflozin and sitagliptin fixed dose combination in patients with type 2 diabetes mellitus in India**

### **Study Design:**

- This is a real-world, retrospective, observational, electronic medical record (EMR) based study
- In this study, EMR data from the patients satisfying the eligibility criteria was retrieved and analyzed

### **Inclusion Criteria:**

- Male and female patients of age  $\geq 18$  years.
- Patients diagnosed with type-2 diabetes mellitus
- Patients with HbA1c  $\geq 7\%$  at baseline visit.
- Patients prescribed dapagliflozin and sitagliptin FDC in any visit other than baseline visit on the EMR platform
- Patients with EMR data captured for more than one visit

### **Exclusion Criteria:**

- Patients diagnosed with type-1 diabetes mellitus.
- Patients are on insulin or any other injectable antidiabetic medication like glucagon-like peptide-1 (GLP -1) agonists.
- Patients with incomplete medical records with respect to the required study parameters

The study retrieved EMRs of patients who met the eligibility criteria and had been prescribed dapagliflozin and sitagliptin fixed-dose combination (FDC) and were at least 18 years old.

Ethics committee approval

Retrieve EMR data

Filter patients meeting eligibility criteria

Visit 1 (Baseline visit)

Visit 2 (3 months  $\pm$  1 month from baseline)

Visit 3 (6 months  $\pm$  1 month from baseline)

Study protocol designed - consent

Demographic characteristics, symptoms, diagnosis, HbA1c, glucose levels, comorbidities, medication details

- The Baseline visit (Visit 1) reading was the visit in which the patient was prescribed with dapagliflozin and sitagliptin FDC as captured in EMR (2022-2023)
- And at follow-up visits (Visit 2 and Visit 3) reading after the baseline visit in this study.

### **Primary Endpoint:**

- Mean change in HbA1c from baseline to 3 months in patients with HbA1c  $\geq 8\%$  at baseline

### **Secondary Endpoint:**

- Mean change in HbA1c from baseline to 3 months in patients with HbA1c  $\geq 8\%$  at baseline, upon stratification of patients by cardiovascular comorbidities

| Patient Disposition (Category)                                                                                                                                                        | Patient count (n) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Total patients in EMR records from 2022-2023                                                                                                                                          | 5213919           |
| Total patients in EMR $\geq 18$ years                                                                                                                                                 | 4667883           |
| Patients in EMR $\geq 18$ years diagnosed with type 2 diabetes mellitus                                                                                                               | <b>11,99, 314</b> |
| EMR records of patients with Dapagliflozin and Sitagliptin fixed-dose combination (FDC)                                                                                               | <b>58,020</b>     |
| EMR records of patients prescribed with Dapagliflozin and Sitagliptin fixed-dose combination (FDC) in any visit other than the baseline visit on EMR platform.                        | 32130             |
| Patients with glycosylated hemoglobin (HbA1c) $\geq 8\%$ at baseline visit prescribed with dapagliflozin + sitagliptin FDC in any visit other than the baseline visit on EMR platform | 8525              |
| Patients with glycosylated hemoglobin (HbA1c) $\geq 7\%$ at baseline visit prescribed with dapagliflozin + sitagliptin FDC in any visit other than the baseline visit on EMR platform | <b>12,060</b>     |
| Patients meeting eligibility (inclusion and exclusion) criteria                                                                                                                       | <b>3112</b>       |

# Demographic and Other Baseline Characteristics

| Parameter                |                                               | Mean Value             |
|--------------------------|-----------------------------------------------|------------------------|
| Age (Years)              |                                               | 54.71 ± 10.6           |
| Weight (Kg)              |                                               | 72.77 ± 12.33          |
| Gender                   | Male                                          | 1699 ( <b>54.5 %</b> ) |
|                          | Female                                        | 1413 ( <b>46.5 %</b> ) |
| BMI (kg/m <sup>2</sup> ) | Overall                                       | <b>28.24</b> ± 4.98    |
|                          | Underweight (<18.50 kg/m <sup>2</sup> )       | 17.76 ± 0.67           |
|                          | Normal (18.50 - 22.99 kg/m <sup>2</sup> )     | 21.72 ± 1.1            |
|                          | Overweight (23.00 - 24.99 kg/m <sup>2</sup> ) | 24.08 ± 0.59           |
|                          | Obese (≥25.00 kg/m <sup>2</sup> )             | 29.9 ± 4.52            |

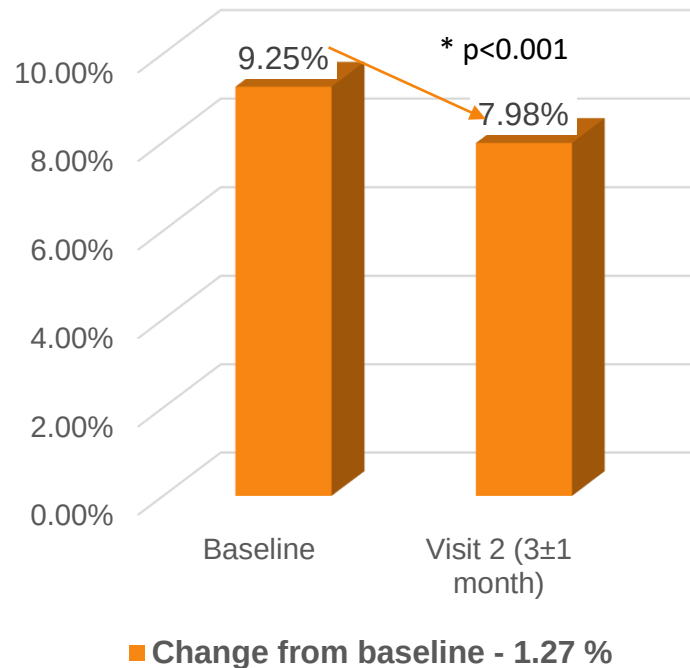
# Results: Change in HbA1c in subset analysis of patients with CV comorbidities

For 640 (76%) patients, CV comorbidities data was available for baseline and Visit-2 (3±1month)

Change from baseline (CFB) at Visit-2 was statistically significant ( $9.25 \pm 1.26\%$  to  $7.98 \pm 1.16$ )

**CFB:  $-1.27 \pm 1.44$ ;  
 $p < 0.001$ .**

**n = 640**  
**Change in HbA1c: at 3 months**



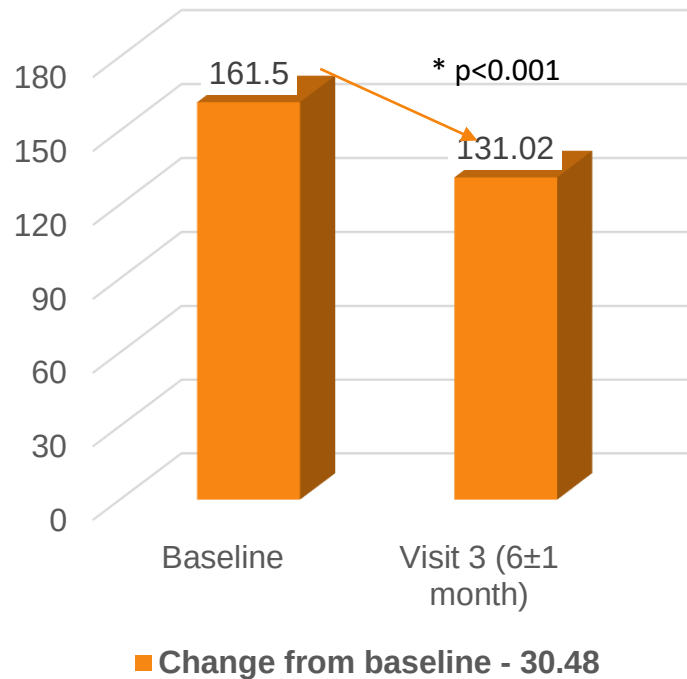
## Results: Change in FBG & PPBG

These 640 patients, FBG data was available for baseline and at Visit-3 (6±1 month)

CFB at Visit-2 was statistically significant (161.50±47.40 to 131.02±32.03)

**CFB -30.48,  
p<0.0001.**

**n = 640**  
**Change in FBG: at 3 months**

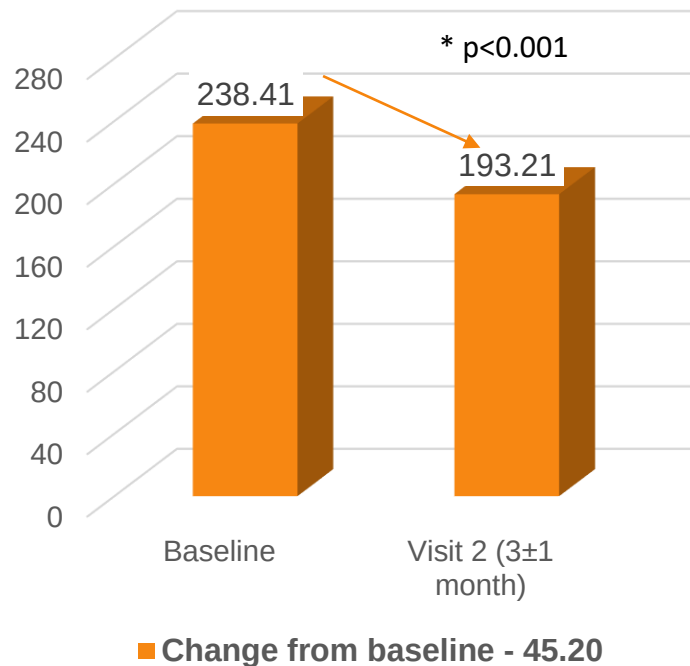


These 640 patients, PPBG data was available for baseline and at Visit-3 (6±1 month)

Change from baseline (CFB) at Visit-2 was statistically significant (238.41±65.38 to 193.21±57.38)

**CFB=-45.20;  
p<0.0001)**

### n = 640 Change in PPBG: at 3 months

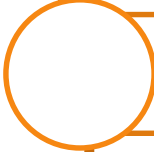


- This a real world EMR based evidence **captured on digital platform**
- The database on which this study was conducted, **safety data was not captured**

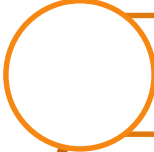
- **ADA 2023:** Adult diabetics patients having  $\geq 1$  risk of ASCVD, HF, DKD or high risk of ASCVD consider SGLT2 inhibitors or GLP- 1 RA
- **RSSDI 2022:** Intensive glycemic control with antidiabetic drugs reduces CV risk and complications in patients with T2DM. SGLT2 inhibitors and GLP-1 analogues provide CV benefits in patients with T2DM



**In patients with T2DM, CV comorbidities and HbA1c $\geq$ 8% at baseline, FDC of Dapagliflozin and Sitagliptin showed significant improvement in HbA1c levels**



Majority of patients receiving dapagliflozin and sitagliptin FDC were having CV comorbidities



This combination therapy is beneficial for achieving glycaemic control in patients with uncontrolled hyperglycaemia, with potential for CV benefits



**To conclude this FDC of Dapagliflozin and Sitagliptin can be considered as part of cardiometabolic risk management in T2DM patients in real world setting**

*Thank you*